

Storing energy and sharing entanglement, read in qg: a charging rule for qubit batteries and a slot rule for Bell-pair distillation

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Abstract

Qubit batteries and Bell pairs stored in network memories both age under the same two processes, relaxation (T1) and dephasing (T2). Both are also judged by a few expectation values: the polar bias $qg_Z = \langle \sigma_z \rangle$ and its companions qg_X, qg_Y [1] for a battery, and three correlations c_x, c_y, c_z with two local polar biases for a pair. We write both problems in these measured quantities and ask what practical rules follow. (i) For a qubit battery the ergotropy is $W = (\omega/2)(r - qg_Z)$, with r the Bloch length. A battery used before the time t_c should be charged fully; after it, it should be tilted, to about 0.6π at $t = T_1$. The crossover solves $x^{2T_1/T_2-1} = 2(2x-1)$ with $x = e^{-t_c/T_1}$. Coherence buys storage time only when T_2 is comparable to T_1 . Certifying the stored work from shots is expensive: the plug-in estimate reports work in a passive state in every run, and a safe bound with 1000 shots per axis certifies 40% of it. A Dicke register stores charge that no single qubit can release, and the register-mean qg_Z cannot see this, while the sum of local ergotropies can. (ii) For a Bell pair, each Pauli noise leaves one correlation at its ideal value, and only memory T1 moves the local polar biases, which gives a T1 witness on a shared pair. The textbook DEJMPS order can lower the fidelity (0.820 to 0.705 under Y-flip noise); putting the smallest error estimated from 200 shots per setting in the combined slot reaches the best output. For BBM92 with stored pairs, the latest usable storage time has a closed form in the aged correlations, and one distillation round roughly doubles it. All results are simulations, reproduced by scripts and pinned by regression tests.

1 Two problems, one description

A qubit battery with Hamiltonian $H = \omega|1\rangle\langle 1|$ and a Bell pair waiting in two memories age under the same channels. Amplitude damping with $g = 1 - e^{-t/T_1}$ moves populations towards $|0\rangle$, and dephasing shrinks coherences as e^{-t/T_2} . The questions differ: how much work can still be extracted, and whether a pair is still good enough to distil or to use for a key. In both cases the answer depends on a handful of expectation values, and those are the numbers an experiment measures. This note collects what the qg description of these values gives in practice, and states which parts are known.

2 Qubit batteries

Ergotropy in qg. A qubit with Bloch vector (qg_X, qg_Y, qg_Z) of length r stores energy $\omega(1 - qg_Z)/2$. Its ergotropy [2], the work a unitary can extract, is

$$W = \frac{\omega}{2}(r - qg_Z) = W_{\text{inc}} + W_{\text{coh}}, \quad W_{\text{inc}} = \omega \max(0, -qg_Z), \quad W_{\text{coh}} = \frac{\omega}{2}(r - |qg_Z|). \quad (1)$$

Table 1: Qubit battery storage. Best angle and ergotropy (best / fully inverted / equator), in units of ω .

T_2/T_1	t_c	$t = 0.5 T_1$	$t = T_1$	$t = 2 T_1$
2	$0.288 T_1 = \ln(4/3)$	$0.70\pi: .297 / .213 / .240$	$0.58\pi: .129 / 0 / .122$	$.038 / 0 / .038$
1	$0.405 T_1 = \ln(3/2)$	$0.78\pi: .234 / .213 / .165$	$0.59\pi: .054 / 0 / .050$	$.005 / 0 / .005$
0.5	$0.618 T_1$	$\pi: .213 / .213 / .073$	$0.60\pi: .008 / 0 / .007$	≈ 0
0.2	$0.692 T_1$	$\pi: .213 / .213 / .004$	≈ 0	≈ 0

It agrees with the general eigenvalue formula to 3×10^{-16} on 200 random states, and the storage curves below agree with an integrated Lindblad equation.

Storage and the charging rule. A battery charged at polar angle θ ages as $qg_Z(t) = 1 - (1 - \cos \theta)e^{-t/T_1}$ and $r_\perp(t) = \sin \theta e^{-t/T_2}$. Table 1 gives the best angle and the ergotropy for three strategies. Full charging ($\theta = \pi$) is best if the battery will be used before a crossover time t_c . After t_c a tilted battery stores more, because the inverted one stops storing anything at $t = T_1 \ln 2$, while the coherent part decays with T_2 . A small-tilt analysis gives

$$x^{2T_1/T_2-1} = 2(2x - 1), \quad x = e^{-t_c/T_1}, \quad (2)$$

which matches the numerical optimum to 10^{-6} and tends to $T_1 \ln 2$ as T_2 shrinks.

The practical reading is short. Charge fully if the battery will be used before t_c ; otherwise tilt it, to about 0.6π at $t = T_1$. At $t = T_1$ the tilted battery stores 0.13ω for $T_2 = 2T_1$ and 0.05ω for $T_2 = T_1$, where the inverted one stores nothing. At $T_2 = 0.2T_1$ coherence buys nothing. Platforms without relevant T_1 on the storage time scale, such as trapped-ion qubits, should always charge fully.

Certifying the stored work. Eq. (1) needs r , and r comes from three estimated components. Shot noise in each inflates r , so the plug-in estimate overestimates W in more than half of the runs and reports positive ergotropy for a passive state in every run. A 3σ lower bound is never unsafe, but it is costly: with 1000 shots per axis it certifies 0.086 of 0.213ω , and nothing for a nearly discharged battery.

Locked charge in registers. For $n = 4$ qubits with two excitations and $T_2 = T_1$, the Dicke state $D(4, 2)$ stores 2ω , all of it globally extractable, but each qubit alone is passive ($qg_Z = 0$, no coherence). The sum of local ergotropies is 0 for the Dicke battery and 2ω for a product battery with the same energy. The register mean of qg_Z is the same for both and cannot tell them apart; the local sum can. The locked charge is also more fragile under local noise: 0.71 vs 1.05ω at $0.3T_1$, and 0.05 vs 0.41ω at $0.7T_1$.

3 Bell pairs in a network

Five numbers. A pair meant to be $|\Phi^+\rangle$ is described by $c_x = \langle XX \rangle$, $c_y = \langle YY \rangle$, $c_z = \langle ZZ \rangle$ and the local biases $m_A = \langle Z_A \rangle$, $m_B = \langle Z_B \rangle$. Three measurement settings give all five. The fidelity is exact for any state,

$$F = \frac{1}{4} (1 + c_x - c_y + c_z), \quad (3)$$

and the Bell-diagonal weights of the twirled pair, $\Phi^-: (1 - c_x + c_y + c_z)/4$, $\Psi^+: (1 + c_x + c_y - c_z)/4$, $\Psi^-: (1 - c_x - c_y - c_z)/4$, say which error dominates: Φ^- is a phase flip, Ψ^+ a bit flip, Ψ^- both.

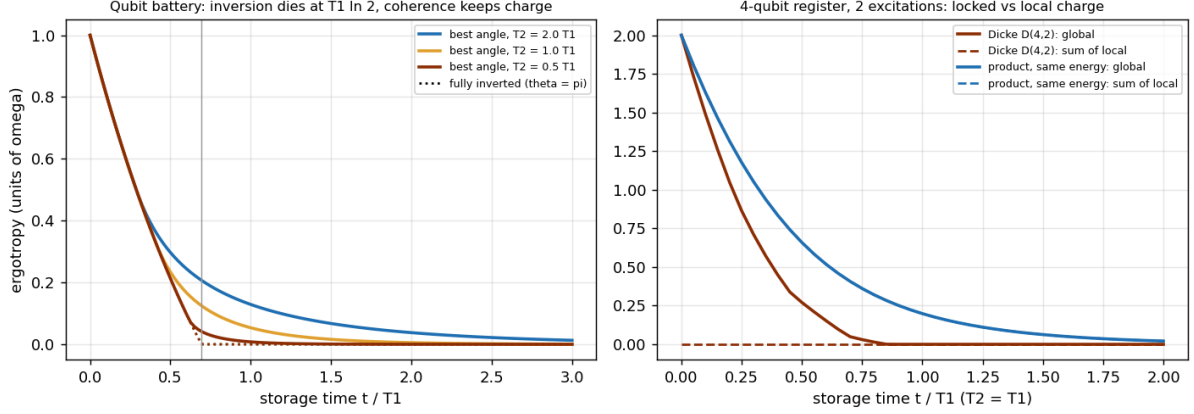


Figure 1: Left: ergotropy at the best charging angle against storage time for three values of T_2/T_1 ; the fully inverted battery is empty after $T_1 \ln 2$ (vertical line). Right: a 4-qubit register with two excitations ($T_2 = T_1$): global ergotropy and sum of local ergotropies for $D(4, 2)$ and for a product battery with the same energy.

Table 2: Noise signatures of a stored pair ($p = 0.1$ on each half; amplitude damping $g = 0.1$).

noise	c_x	c_y	c_z	$m_A = m_B$	F	largest error
dephasing	+0.64	-0.64	+1.00	0	0.820	Φ^-
bit flip	+1.00	-0.64	+0.64	0	0.820	Ψ^+
Y flip	+0.64	-1.00	+0.64	0	0.820	Ψ^-
depolarizing	+0.81	-0.81	+0.81	0	0.858	all equal
amplitude damping	+0.90	-0.90	+0.82	+0.10	0.905	Ψ^+

Noise signatures. Table 2 shows the five numbers for $p = 0.1$ on each half. Each Pauli noise keeps one correlation at its ideal value. Only amplitude damping, that is memory T1, moves m_A and m_B away from zero. This is the T1 witness used for BB84 in the companion note on cryptography, now on a shared pair, and it is read from the same ZZ shots.

Which distillation. The DEJMPS recurrence [3] (a refinement of Ref. [4]) combines the Φ^+ weight A with the error weight B that sits in one slot, $A' = (A^2 + B^2)/N$, and local rotations choose which error goes there. Table 3 compares the textbook slot (Ψ^-), the best slot, and a rule applied from data: put the smallest error estimated from the three correlations, with 200 shots per setting, in slot B .

The fixed order fails when the dominant error sits in the slot combined with Φ^+ . For Y-flip noise one round lowers the fidelity, from 0.820 to 0.705, and for amplitude damping it gains almost nothing (0.828 against a possible 0.920). The data-driven rule reaches the best output in every case, 94% of the time for amplitude damping, where two error weights are close.

Memory cutoff for BBM92. For entanglement-based key distribution [5] with $\text{QBER}_Z = (1 - c_z)/2$, $\text{QBER}_X = (1 - c_x)/2$ and secret fraction $1 - h(\text{QBER}_X) - h(\text{QBER}_Z)$, the aged correlations have closed forms,

$$c_x = -c_y = e^{-2t/T_2}, \quad c_z = 1 - 2g + 2g^2, \quad m = g, \quad g = 1 - e^{-t/T_1}, \quad (4)$$

so the latest usable storage time follows directly (Table 4). One distillation round with the best slot roughly doubles it when T1 limits storage. It costs half the pairs, so per raw pair it pays only after the crossover time in the last row.

Table 3: One DEJMPS round: output fidelity (success probability).

input	F_{in}	textbook slot	best slot	qg rule, 200 shots/setting
dephasing 0.1	0.820	0.954 (0.705)	0.954 (0.705)	0.954 (right slot 100%)
bit flip 0.1	0.820	0.954 (0.705)	0.954 (0.705)	0.954 (100%)
Y flip 0.1	0.820	0.705 (1.000)	0.954 (0.705)	0.954 (100%)
depolarizing 0.1	0.858	0.891 (0.828)	0.891 (0.828)	0.891 (100%)
amplitude damping 0.2	0.820	0.828 (0.820)	0.920 (0.731)	0.915 (94%)

Table 4: Latest storage time with a positive BBM92 secret fraction, in units of T_1 .

T_2/T_1	2	1	0.5	0.2
no distillation	0.184	0.129	0.088	0.051
one DEJMPS round	0.395	0.232	0.144	0.076
distilling pays per raw pair after	0.085	0.083	0.056	0.032

4 What is common

Three points hold in both settings. First, T1 and T2 play different roles, and the ratio T_2/T_1 decides the rule: how far to tilt a battery, and how much a distillation round extends a pair’s life. Second, a local polar bias is a free T1 witness: qg_Z of a stored battery and m_A, m_B of a stored pair come from shots already taken. Third, single-qubit or register-mean quantities miss what is shared: the register mean of qg_Z cannot see locked charge, and local biases cannot see which Bell error dominates. The useful description keeps the local numbers and the few correlations that matter.

5 Limitations

(i) Everything is simulation with independent Markovian T1 and T2 channels and perfect local gates; no hardware data. (ii) Eq. (1), coherence-assisted storage, locally passive but globally charged batteries [6], Eq. (3), the Bell-diagonal reading, DEJMPS with local rotations and memory cutoffs are known. The contributions are the charging rule with its closed-form crossover Eq. (2), the certification cost, the qg reading of which charge is locally extractable, the noise signatures with the T1 witness on a pair, the slot rule applied from finite-shot data, and the closed-form cutoff. (iii) Distillation uses the twirled state. (iv) The BBM92 secret fraction is asymptotic; finite-key effects would shorten every cutoff in Table 4. (v) Charging and extraction are ideal unitaries; their cost and duration are not modelled.

6 Code availability

examples/battery_ergotropy_qg.py and examples/bell_pairs_network_qg.py in <https://github.com/Viny2030/qang> (MIT license), with regression tests. Sections 56 and 57 of RESEARCH_NOTES.md give more detail.

References

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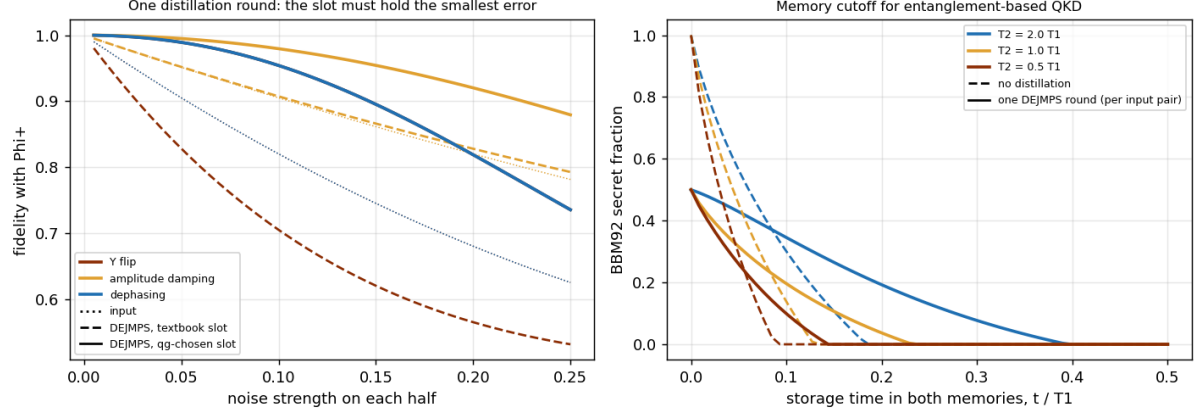


Figure 2: Left: fidelity after one DEJMPS round against noise strength, with the textbook slot (dashed) and the slot chosen from qg data (solid); dotted lines are the input. Right: BBM92 secret fraction against storage time, without distillation (dashed) and after one round, per input pair (solid).

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